

 Marwadi University Marwadi Chandarana Group	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: To enhance five degraded grayscale images using suitable gray-level transformation techniques and observe how each transformation improves the visual quality of the image.	
OPED-7	Date:15-08-2026	Enrollment No:92400133030

Gray-Level Transformations

Contrast Stretching

Purpose

Contrast stretching is used when an image has a **small range of gray levels**.

It expands the intensity range so that dark pixels become darker and bright pixels become brighter.

Formula

$$s = (r_{max} - r_{min}) / (r - r_{min}) \times 255$$

Where:

- r = input pixel
- r_{min} = minimum intensity
- r_{max} = maximum intensity
- s = output pixel

Observation

The first image in the assignment appears **gray and low-contrast**. Contrast stretching increases the difference between dark and bright regions.

Result: Objects become more clearly visible and the overall image looks sharper.

Histogram Equalization

Purpose

Histogram Equalization improves the contrast of an image by redistributing pixel intensity values.

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It is useful when the image has poor contrast.

Code

```
def histogram_equalization(image):  
    return cv2.equalizeHist(image)
```

Observation

The second image shown in the assignment contains a **medical/CT-like circular region with noise**.

Histogram equalization can improve the visibility of different intensity regions.

However, normal histogram equalization may also make the noise more noticeable.

Therefore, **CLAHE** can be used for better local contrast.

Gamma Transformation

Gamma transformation is useful for **very dark or very bright images**.

The formula is:

For a dark image:

$\text{gamma} < 1$

is normally used to make the image brighter.

Code:-

```
def gamma_correction(image, gamma):  
    normalized = image / 255.0  
    corrected = np.power(normalized, gamma)  
    corrected = corrected * 255
```

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```
return corrected.astype(np.uint8)
```

Observation

The third image in the assignment is very dark.

Gamma transformation with $\gamma < 1$ increases the intensity of dark pixels.

Result:

Dark objects become visible.

Hidden details can be observed.

Overall brightness increases.

Log Transformation

Log transformation is particularly useful for images where important information is concentrated in **low-intensity values**.

Formula

$$s = c \log(1 + r)$$

Observation

The fourth image is extremely dark, with only a small bright region visible.

Log transformation expands the low-intensity values and makes dark details more visible.

Result:

- Dark areas become brighter.
- Hidden structures become easier to observe.
- Small intensity differences become more noticeable.

Observation

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The fourth image is extremely dark, with only a small bright region visible.

Log transformation expands the low-intensity values and makes dark details more visible.

Result:

- Dark areas become brighter.
- Hidden structures become easier to observe.
- Small intensity differences become more noticeable.

Negative Transformation

Another useful gray-level transformation is image negative.

Formula

$$s = L - 1 - r$$

For an 8-bit image:

$$s = 255 - r$$

Observation

The fifth image is very dark, so the negative transformation can make structures that are difficult to see in the original image more noticeable.

Result:

- Dark areas become bright.
- Bright areas become dark.
- Some hidden structures become easier to inspect.

Gray-Level Slicing

Gray-level slicing is used when we want to highlight a **specific range of intensity values**.

For example, suppose we want to highlight pixels between 150 and 255.

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Observation

The fourth image contains a small bright object against a very dark background.

Gray-level slicing can be used to isolate bright intensity regions.

Result:

- Bright object becomes clearly visible.
- Background is suppressed.
- Important high-intensity regions can be isolated.

Code:-

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# 1. Contrast Stretching
def contrast_stretching(image):
    min_val = np.min(image)
    max_val = np.max(image)
    if max_val == min_val:
        return image
    result = ((image.astype(np.float32) - min_val) /
              (max_val - min_val)) * 255
    return result.astype(np.uint8)

# 2. CLAHE Enhancement
def clahe_enhancement(image):
    clahe = cv2.createCLAHE(
```

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```

clipLimit=2.0,
tileGridSize=(8, 8)
)
return clahe.apply(image)

# 3. Gamma Correction
def gamma_correction(image, gamma):
    normalized = image.astype(np.float32) / 255.0
    result = np.power(normalized, gamma)
    result = result * 255
    return result.astype(np.uint8)

# 4. Log Transformation
def log_transformation(image):
    image_float = image.astype(np.float32)
    maximum = np.max(image_float)
    if maximum == 0:
        return image
    c = 255 / np.log(1 + maximum)
    result = c * np.log(1 + image_float)
    return result.astype(np.uint8)

# 5. Negative Transformation
def negative_transformation(image):
    return 255 - image

image_names = [
    "IMG_1.jpg",

```

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```

"IMG_2.jpg",
"IMG_3.jpg",
"IMG_4.jpg",
"IMG_5.jpg"
]
for i, name in enumerate(image_names):
    image = cv2.imread(name, cv2.IMREAD_GRAYSCALE)
    if image is None:
        print("ERROR: Cannot find image:", name)
        print("Please check the image name and location.")
        print()
        continue
    if i == 0:
        enhanced = contrast_stretching(image)
        technique = "Contrast Stretching"
    elif i == 1:
        enhanced = clahe_enhancement(image)
        technique = "CLAHE"
    elif i == 2:
        enhanced = gamma_correction(image, 0.4)
        technique = "Gamma Correction"
    elif i == 3:
        enhanced = log_transformation(image)
        technique = "Log Transformation"

```

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else:

enhanced = negative_transformation(image)

technique = "Negative Transformation"

plt.figure(figsize=(10, 5))

Original image

plt.subplot(1, 2, 1)

plt.imshow(image, cmap="gray")

plt.title("Original Image")

plt.axis("off")

Enhanced image

plt.subplot(1, 2, 2)

plt.imshow(enhanced, cmap="gray")

plt.title(technique)

plt.axis("off")

plt.tight_layout()

plt.show()

output_name = "enhanced_" + str(i + 1) + ".jpg"

success = cv2.imwrite(output_name, enhanced)

if success:

print("-----")

print("Image :", name)

print("Technique :", technique)

print("Output :", output_name)

print("-----")

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print()

else:

print("ERROR: Could not save", output_name)

OUTPUT:-

Original Image



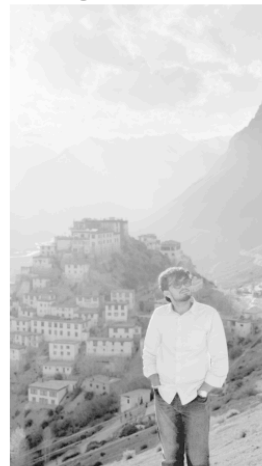
Contrast Stretching



Original Image



Log Transformation



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Original Image



CLAHE



Original Image



Gamma Correction



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Original Image



Negative Transformation

